# TEXMACS in 60 minutes

by Andreas Seidl

Updated for TEXMACS 2.1.5 in 2026.

This document will enable you to quickly understand what you can do with TEXMACS, how you can install it, and how to get started using it. For the official source of information see www.texmacs.org.

**Step 1.** Features and current issues (section 1).

**Step 2.** Download and install (section 2).

**Step 3.** Hands-on tutorial (section 3).

## 1 Features and current issues

To know about the features and issues of a software plays a key role in deciding whether it is sensible to start working with it: are the bugs small and the workarounds painless enough to justify the benefits?

### 1.1 Features

The following is an incomplete list of features, just to give an idea of what you can do with TEXMACS.

- **Creating and editing structured text.**
  - The usual constructs from LATEX: title, author, abstract, sections, are available.
  - The same holds for environments like tables and figures, as well as for mathematical environments like definition, theorem, proof.
  - Mathematical content can be input as known from LATEX (type `\alpha`, `\frac`, `\sum`...), with keyboard shortcuts, with the menus and with the mathematical toolbar. What you see on the screen is the typeset formula.
  - Inclusion of images (Pdf, Svg, Png, Jpeg..., and Eps or PostScript when Ghostscript is installed), and drawing with the integrated graphics editor (Insert→Image→Draw image).
  - Labels for sections, equations, theorems, tables and figures, and references to them, also to their page numbers.
  - Citations and bibliographies based on BibTEX files (an integrated BibTEX engine is included).
  - Automatically generated (and updated) table of contents, bibliography, index, glossary.
  - Laptop presentations (style `beamer`), posters, letters, books.
- **Importing and exporting** (File→Import and File→Export).
  - LATEX (import and export, also import of LATEX classes)
  - Html
  - TEXMACS Xml
  - BibTEX
  - TEXMACS Scheme, verbatim text and source code
  - Pdf: export, also with the TEXMACS document embedded in the Pdf; such a Pdf can be imported back (File→Import→Pdf with embedded document)
  - PostScript (export only); a selection can also be exported as an image.

- **Plug-ins: interfaces to external programs.** With Insert→Session you open an interactive session inside your document, whose results are typeset by TEXMACS. There are plug-ins for computer algebra systems (MAXIMA, GIAC, FRICAS, REDUCE, SAGE, MATHEMATICA, MAPLE, PARI, MACAULAY2...), numerical systems (OCTAVE, SCILAB, R...), programming languages (PYTHON, SCHEME, the SHELL...) and graphics tools (GNUPLOT, ASYMPTOTE, GRAPHVIZ, TIKZ...). The program itself has to be installed on your computer (TIKZ needs a LATEX installation): TEXMACS only shows the plug-ins which it finds.
- **Customization.** Macros, style files and the SCHEME extension language make it possible to adapt TEXMACS to your needs.

### 1.2 Current issues

TEXMACS runs natively on WINDOWS, MACOS and LINUX: the issues of the CYGWIN versions of 2004 (X server, keyboard, clipboard, image converters) are gone. The remaining limitations are mostly the following:

- The plug-ins need the corresponding programs: install for instance MAXIMA or GNUPLOT before using them from TEXMACS.
- The conversions to and from LATEX and HTML work well for usual documents, but complex LATEX macros or packages may need some manual work after an import.
- Some image formats need an external program to be displayed: GHOSTSCRIPT for EPS and POSTSCRIPT, a converter for XFIG.

Problems with the official TEXMACS are reported to its developers, see www.texmacs.org (mailing lists and bug tracker); problems with the TEXMACS Fixes builds go to the issues of github.com/mgubi/texmacs, not to the official TEXMACS. Please include the version of TEXMACS (shown by `texmacs --version`, or on MACOS by the *About* entry of the application menu) and of your operating system.

## 2 Download and installation

Installing TEXMACS takes three steps:

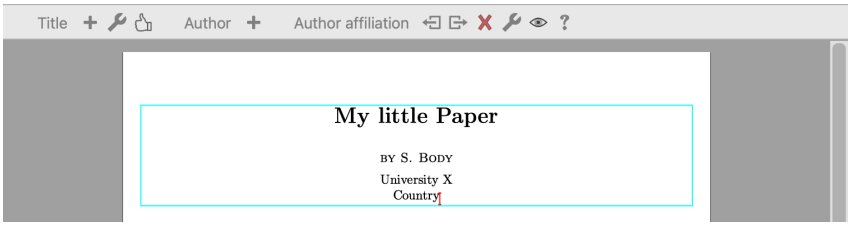
1. **Download.** Ready-made packages of TEXMACS for WINDOWS, MACOS and the usual LINUX distributions are available from the download page of www.texmacs.org. The TEXMACS Fixes version used in this tutorial, with the corrections collected since the last official release, is available as packages for WINDOWS, MACOS and LINUX from mgubi.github.io/texmacs/fixes.html (the files of the pre-release `fixes-latest` of github.com/mgubi/texmacs).
2. **Install.** Official packages: on WINDOWS, run the installer; on MACOS, open the disk image and drag the application to the `Applications` folder; on LINUX, install the package with the package manager of your distribution. TEXMACS Fixes builds: on WINDOWS, run the installer, or unpack the zip archive and start TEXMACS from the unpacked folder; on LINUX, the build is an APPIMAGE: make it executable (`chmod +x` on the file) and run it; on MACOS, the application is not notarized, so the first time open it with a right-click (or Control-click) on it and *Open*, or remove the quarantine flag with `xattr -dr com.apple.quarantine` followed by the path of the application. Nothing else is needed: no CYGWIN, no X server, no TEX installation.
3. **Start** TEXMACS like any other application. The first time, a *Welcome* document (Help→Welcome) shows links to the documentation.

## 3 Hands-on tutorial

The keyboard shortcuts shown below are those of the system on which this document was typeset; the menus show the shortcuts of your own system next to their entries. Instead of the menus, you may also use the icons of the toolbars; the third toolbar, the *focus toolbar*, always shows what can be done with the structure under the cursor.

## 3.1 Part 1: Title, author, abstract, sections and the style of a document

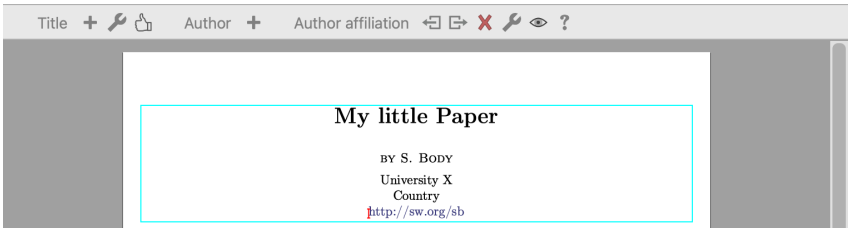
1. Start TEXMACS. Select Insert→Title→Insert title from the menu.
2. Type "My little Paper" and hit ↩.
3. Type some name and hit ↩.
4. Type in some address, with ↩ between the lines. Meanwhile your document should look like:



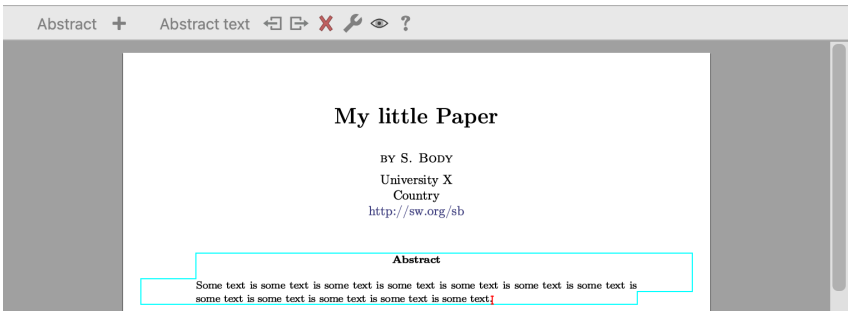

5. Hit ↩ and choose Insert→Link→Hyperlink from the menu. Type in a URL, then hit the → key and retype it. You should see something like:



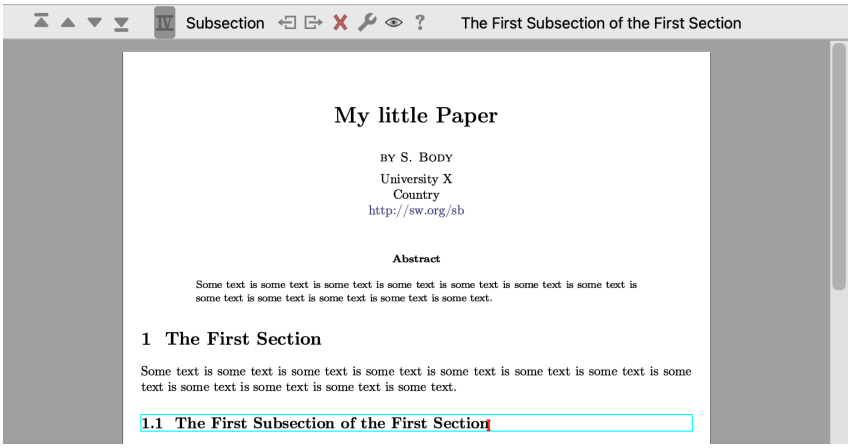

6. Hit ↩ to finish the definition of the hyperlink:



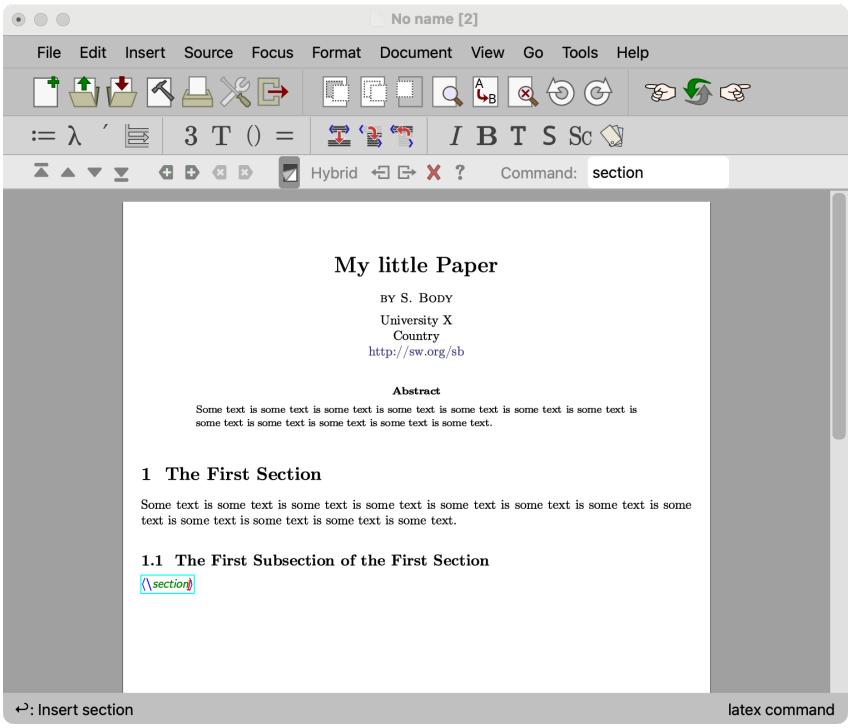

7. Hit ↘ and then the → key: you leave the address and the whole title block. (The cyan box shows the structure in which the cursor is; the focus toolbar and the right side of the footer show it too.)
8. Choose Insert→Title→Abstract from the menu. Enter some text.




9. Hit the → key, and notice how you leave the abstract. Hit ↩, this brings you to the beginning of a new paragraph.
10. Choose Insert→Section→Section from the menu (or type ^1). Type "The First Section". Hit ↩, type some text and hit ↩ again.
11. Choose Insert→Section→Subsection from the menu (or ^2). Type "The First Subsection of the First Section". You see now something like:




12. The sections and subsections are numbered: a new document has the `generic` style. Select Document→Style→article from the menu: the title and the sections take the look of an article.
13. Hit ↩ to get to a new line. There is a second (LaTeX-style) way to enter sections. Type "`\section`". You see:




14. Hit ↩. Now you have achieved the same as just before by selecting Insert→Section→Section from the menu, but without removing your hands from the keyboard! Type in "The Second Section" and hit ↩. Type some text.
15. Save your file with File→Save (⌘S), for instance as `hot1.tm`.

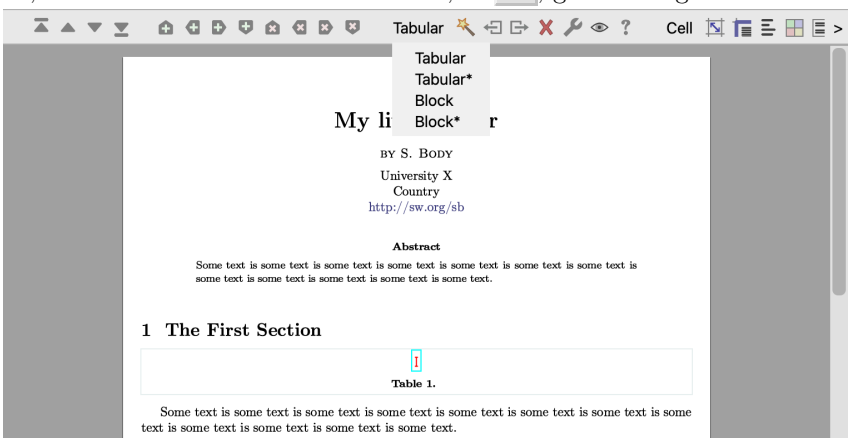
This finishes the first part of the hands-on tutorial.

## 3.2 Part 2: Tables and figures

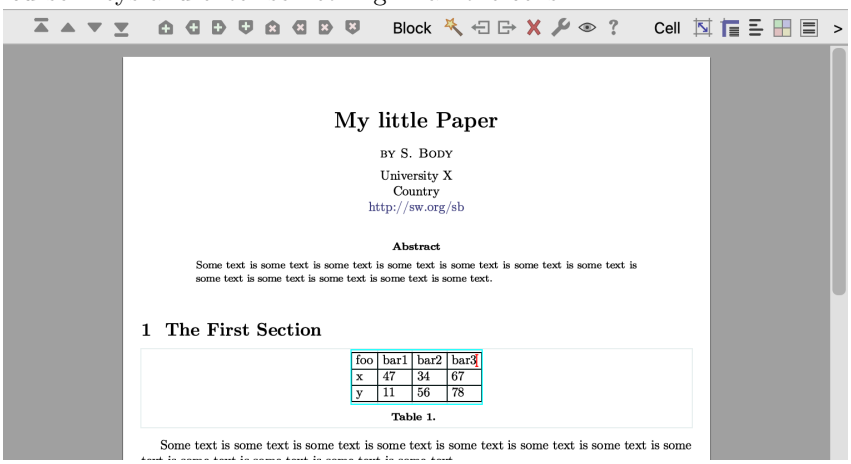
Proceed with your file from part 1.

1. Put the cursor at the end of the heading of the first section and hit ↩ to create a new line after it.
2. Select Insert→Table→Big table from the menu. (There is an icon ▦ on the second toolbar as well, it says "Insert a table" if you move the mouse over it.) This inserts a numbered table with a caption, containing a table without borders (a *tabular*). To get the cell borders, click

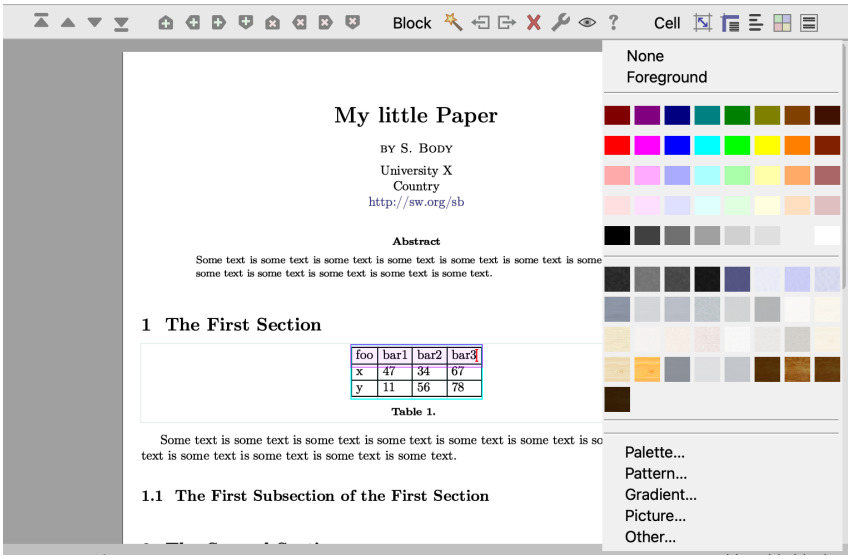
on the name *Tabular* in the focus toolbar (or use Focus→Tabular) and choose the variant *Block*; the wand icon of the focus toolbar, or ⌥→|, goes through the variants too.



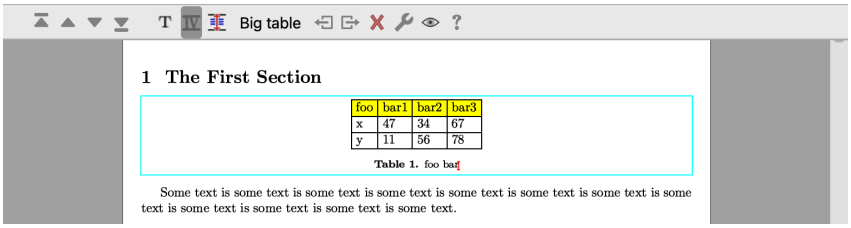

3. By pressing ^→ three times, you get three more columns. Accordingly, by pressing ^↓ twice you get two more rows. On MACOS, these shortcuts are taken by default by MISSION CONTROL: use instead the icons of the focus toolbar which insert a column to the right and a row below (the round icons with a cross, at the left of the focus toolbar). Navigate with the cursor keys and enter something in all the cells.



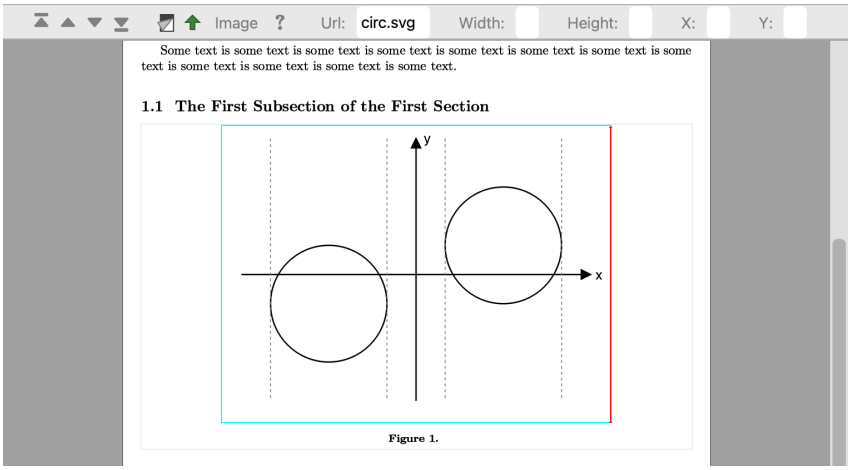

4. Mark the four topmost cells by clicking inside the leftmost cell, holding the mouse button pressed, and moving to the rightmost cell (works as well in the other direction). Then click on the icon of the focus toolbar ("Set background color of cell") and pick yellow:



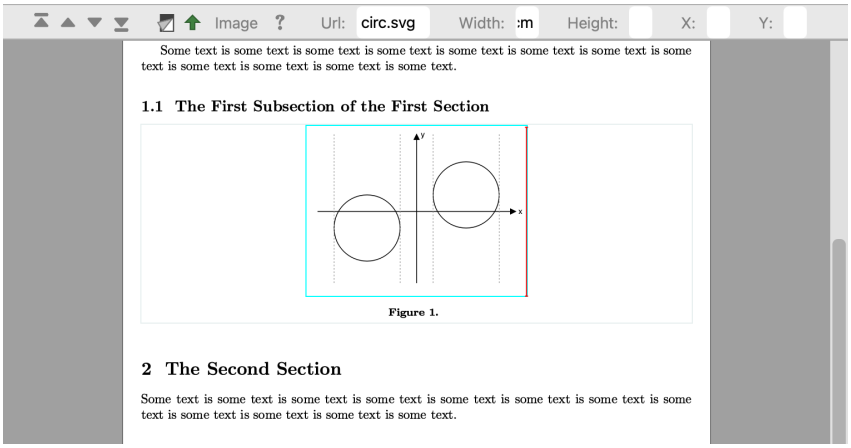
5. Click after “Table 1.” and enter some caption. The result looks like:




6. Create a new line after the heading of the first subsection.
7. Select Insert→Image→Big figure, and then Insert→Image→Insert image from the menu. (There is an icon as well, it says “Insert a picture” if you move the mouse over it.) You get a dialog, in which you choose an image file: PDF, SVG, PNG, JPEG, GIF... are all accepted (EPS too if GHOSTSCRIPT is installed), and the dialog shows a preview. Select any image (we used a drawing of two circles, `circ.svg`):




8. This is too big. To edit the size, look at the focus toolbar while the cursor is just after the image: it shows the file name and the fields *Width* and *Height*. Type for instance `6cm` in the *Width* field (or a relative width like `0.4par`, 40% of the width of a paragraph) and hit ↩. The image is smaller now:




9. Save your file as `hot2.tm` (File→Save as).

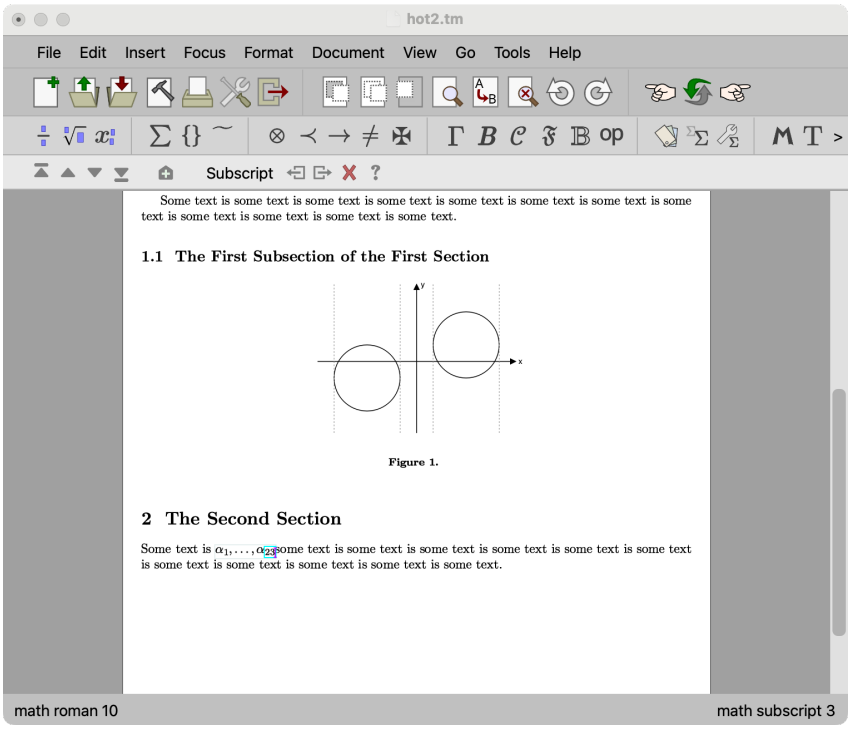
This finishes the second part of the hands-on tutorial.

## 3.3 Part 3: Mathematical content

Proceed with your file from part 2.

1. Go somewhere in the text of section 2 of the document and type: `$\alpha_1→,\ldots↩,\alpha_23`. (A LaTeX-like command is confirmed with ↩, or with

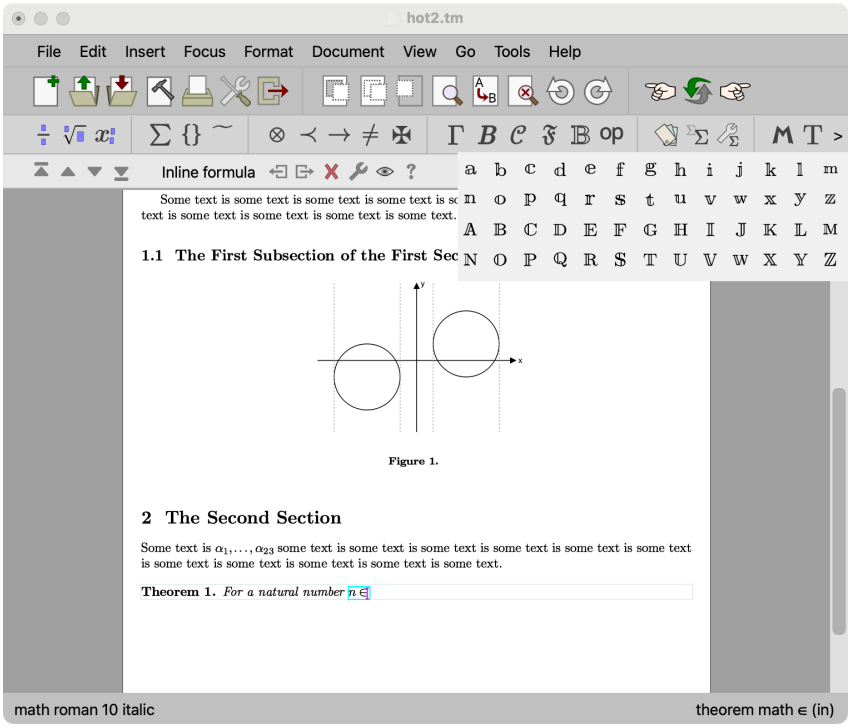
␣, _ or ^, which then also do their usual job.) While you are inside the formula, the mathematical toolbar replaces the text toolbar:



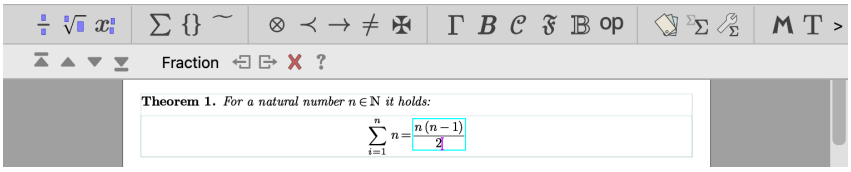

2. Type → to leave the index, → again to leave the formula, and hit ␣ to insert a space before the word “some”.

3. Go to the end of the paragraph and hit ↩ to make a new line.

4. Type: `\theorem`↩`For a natural number $n\in`↩ and select from the icon 𝔹 of the mathematical toolbar the blackboard bold version of N (or type ⇧N ⇧N, i.e. two capital N):



5. Type → to leave the formula, then type: "it holds:"
6. Now we want to insert a displayed formula. Select Insert→Mathematics→Displayed formula from the menu (or faster, hit ^$).
7. Type: \sum↩_i=1→^n↘n=\frac↩n*(n-1)↓2. Notice that "*" results in a little space and not in an explicit multiplication sign (it is an invisible multiplication). If you press →| after "*", then the visible symbol is chosen. Typing "(" inserts both brackets.



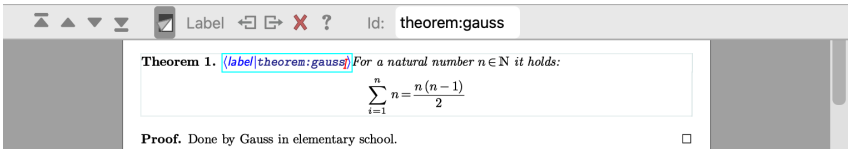

8. Press → three times and then ↩: you leave the formula and the theorem, and get to a new line.
9. Type: \proof↩Done by Gauss in elementary school.
10. Save your file as hot3.tm.

This finishes the third part of the hands-on tutorial.

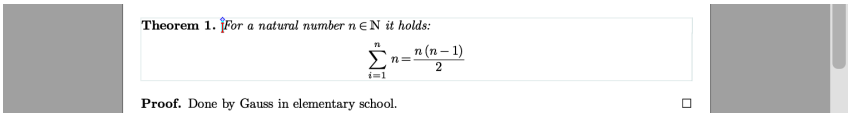
## 3.4 Part 4: References

Proceed with your file from part 3.

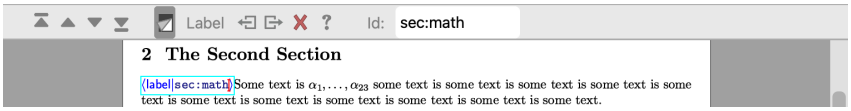
1. Move the cursor just before the text "For a natural number", and enter: \label↩theorem:gauss (or use Insert→Link→Label, ^!).



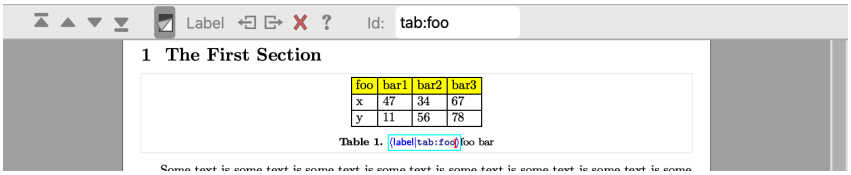

2. Hit ↩ to commit the label. You see a little flag, where the otherwise invisible label sits. This flag won't be printed. If you go behind this flag and hit ⌫, then you can edit the label again (or edit its name in the focus toolbar). Note: the label's name is not important, you could have used "cat" or "dog", but mentioning the environment (theorem, section, etc.) helps later when a reference is made.



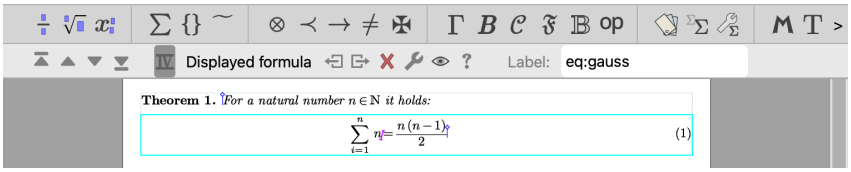

3. Go to the beginning of the first line after the heading of section 2 and type \label↩sec:math. (A label typed inside the heading itself may get a wrong number, because the heading is repeated in the page header.)



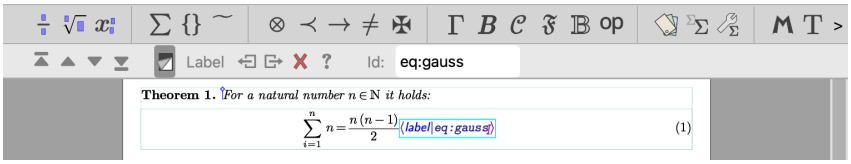

4. Hit ↩ to commit the label.
5. Do a similar thing at the beginning of the caption of the table: \label↩tab:foo, and hit ↩.



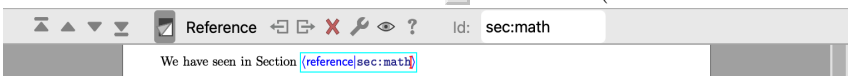
6. Only numbered equations can be referred to. Put the cursor inside the displayed formula, but outside the fraction and the sum (for instance just before the "=" sign), so that the focus toolbar shows *Displayed formula*; then select Focus→Numbered (or the icon of the focus toolbar). The shortcut ⎋# works from anywhere in the formula. The equation gets the number (1):



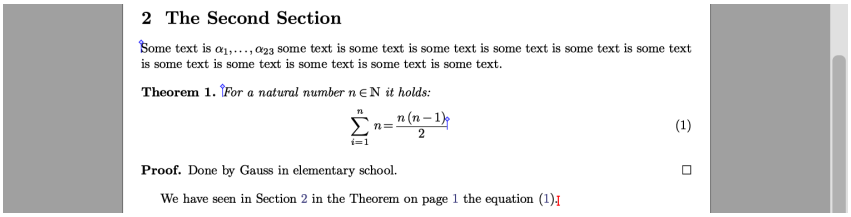

7. Put a label at the end of the equation: `\label`↩`eq:gauss`, and hit ↩.




8. After that many labels, we can finally refer to them! Go, e.g., to the end of the document and write: "We have seen in Section `\ref`↩`sec:math`" (or Insert→Link→Reference):




9. Hit ↩, and go on by typing: "in the Theorem on page `\pageref`↩`theorem:gauss`↩ the equation (`\ref`↩`eq:gauss`↩)." The page numbers are computed when the pages are made: if a page number shows as "?", use Document→Update→All (⌘⇧R). The result is:




10. Play around, insert some bogus theorems, sections and page breaks to see how the labels change automatically!

11. Save your file as `hot4.tm`.

This finishes the fourth part of the hands-on tutorial.

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